Date: _____

Question 5

Input	Output
· choice	· menu
· name	· fine
· ID	· total fine
· code	· borrowed days.
· days	· book. ID.
· type	

Pseudocode:

```
PROCEDURE displayMenu()
    PRINT ("University Menu")
    PRINT ("1. Register No.")
    PRINT ("2. Display Available Books").
    PRINT ("3. Borrow Book")
    PRINT ("4. Return a Book").
    PRINT ("5. Display Record")
    PRINT ("6. Exit")
    PRINT ("Enter Choice").
ENDPROCEDURE
```


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```
PROCEDURE registerMember (~~arr[],~~ name, id, type)
    PRINT ("Enter Name") INPUT name.
    PRINT ("Enter ID") INPUT id.
    PRINT ("Enter Type 1 or 2")
        INPUT type.

    WHILE (type.! = 1 AND type! = 2)
        PRINT ("Invalid").
        INPUT type.

    PRINT ("Member Registered").
ENDRROCEDURE.

PROCEDURE displayBooks (ids[], titles[], copies,
                        bookcount).

    ~~index = 0.~~
    PRINT ("ID", "Titles", "Copies").
    FOR (i = 0, i< bookcount).
        PRINT (ID[index], title[index], 'copies
                                    [index]).

ENDPROCEDURE
```


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```
PROCEDURE borrowBook ( id[], titles[],
        copies[], bookCount$, boorowedIDs[],
        borrowedCount, type)

    enteredID, keepingdays, index, bookIndex = -1.

    IF (type == 1) borrowedlimit = 14
    ELSE borrowedlimit = 30.

    IF (borrowedCount >= 3)
        PRINT ("Borrowing limit Reached").
        RETURN.

    PRINT ("Enter BookID") INPUT enteredID.

    FOR (i = 0, i < bookCount)
        IF (ID[index] == enteredID) bo
            bookIndex = index

    IF (bookIndex == -1).
        PRINT ("Invalid Book").
        RETURN.
```


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```
IF (copies[bookIndex] == 0).
    PRINT ("Sorry No copies Available")
    RETURN

PRINT ("Enter No. of Days")
INPUT keepingDays.

WHILE (keepingDays <= 0).
    PRINT ("Days must be positive").
    INPUT keepingDays

borrowedIDs[borrowedCount++] = enteredID.
copies[bookIndex]--
PRINT ("You have borrowed book", titles[book
        Index], "for", borrowingLimit, "days").

IF (keepingDays > borrowinglimit)
    PRINT ("Fine will be applied")

ENDPROCEDURE.
```

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```
FUNCTION returnBook ( IDs[], copies[], bookCount,
        borrowedIDs[], borrowedCount, type,
            totalFine)

PRINT( " Enter Book ID ") INPUT enteredID

FOR ( i = 0, i < borrowedCount)
    IF ( borrowedIDs[index] == enteredID)
        borrowedIndex = index

IF ( borrowedIndex == -1)
    PRINT ( " You have not borrowed this
                book ").
    Return 0;

PRINT ( " Enter No. of Days ") INPUT keepingDays

WHILE ( keepingDays <= 0)
    PRINT ( " Invalid Days ").
    INPUT keepingDays.

FOR ( i = 0, i < bookCount)
    IF ( Ids[i] = enteredID)
        bookIndex = i
```


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```
copies [bookIndex]++.

IF (type==1)
        borrowing limit =14   , fineRate =20
ELSE    borrowing limit = 30, fine Rate = 10

IF (~~fineRate type~~ ==1).

IF (keepingDays > borrowinglimit).
    extraDays = keepingDays - borrowing limit.
    fineAmant = extradays * fine rate.

    PRINT ("Book Returned late", extraDays, "
            Late fine", fineAmant").
    totalFine += fineAmant

    PRINT ("Enter Membership Code:")
    INPUT & enterCode

    IF (correctCode[i] = & AND enteredCode[i])
        fineAmount /=2
        PRINT ("50% $ waiver")
    ELSE. PRINT ("Incorrect Code").
    PRINT ("Fine:", fineAmant).
ELSE PRINT ("Book Returned on time")
```


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```
    FOR ( i = borrowedIndex ; index < borrowedCount - 1).
        borrowedID[index] = borrowedIDs[index + 1]
    borrowedCount --.
    RETURN FineAmount.
ENDFUNCTION.

PROCEDURE. displayMemberRecord (name[], id,
        type, borrowedIDs[], count, totalfine, fine).

    PRINT ( "Member Record").
    PRINT ( "Name", name).
    PRINT ( "ID ", id).
    PRINT ( "Membership Type" type)
    PRINT ( "Borrowed Books").
    IF (count == 0).
        PRINT ("None")
    ELSE
        FOR ( i = 0, i < count)
            PRINT( borrowedIDs [count).

    PRINT ("Total Fine", total).
    PRINT ( " Fine ", fine)
ENDPROCEDURE
```


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IDs[4] = [101, 102, 103, 104].
DECLARE titles[][] = ARRAY.
DECLARE copies
book count = 4. isregistered = 0.

REPEAT.
displayMenu().
INPUT MenuChoice.

Case of MenuChoices.
case 1:
IF (isRegistered)
PRINT "Member Registered"
ELSE registerMember(name, id, type).
is registered = 1

case 2:
displayBooks(IDs, bitles, copies, book Cant)

case 3:
IF (isRegistered) == FALSE
PRINT ("Register Fist").
ELSE.
borrowbook (ID, titles, copies, book Card
borrowedID, cant, Type)

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```
case 4:
    if ($unregistered == 01)
        PRINT 'Register First'.
    ELSE.
        payableFine += returnBook(IDs, copies, cont,
            borrowedIDs, count, Type, Total).

case 5.
    displayMemberRecord(Name, ID, Type,
        borrowedID, Count, Total, payableFine).

case 6:
    PRINT ('Existing Fine').

UNTIL (menuchoice != 6).
```

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